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<!DOCTYPE html>
<html lang="ko">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>줄의 길이 및 기준 진동수 조절 음높이 탐구 도구</title>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <style>
    :root {
      --primary-color: #2563eb;
      --primary-hover: #1d4ed8;
      --bg-color: #f8fafc;
      --card-bg: #ffffff;
      --text-color: #1e293b;
      --border-color: #e2e8f0;
      --accent-red: #ef4444;
      --accent-green: #10b981;
      --accent-purple: #8b5cf6;
    }

    body {
      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif;
      background-color: var(--bg-color);
      color: var(--text-color);
      margin: 0;
      padding: 16px;
    }

    .container {
      max-width: 1000px;
      margin: 0 auto;
      display: flex;
      flex-direction: column;
      gap: 16px;
    }

    header, .card {
      background: var(--card-bg);
      padding: 18px;
      border-radius: 12px;
      border: 1px solid var(--border-color);
      box-shadow: 0 1px 3px rgba(0,0,0,0.05);
    }

    h1 { margin: 0 0 8px 0; font-size: 1.4rem; color: var(--primary-color); }
    .grid-layout { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; }
    @media (max-width: 768px) { .grid-layout { grid-template-columns: 1fr; } }
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.manual-table-container {
  max-height: 220px;
  overflow-y: auto;
  border: 1px solid var(--border-color);
  border-radius: 6px;
  margin-bottom: 10px;
}

table { width: 100%; border-collapse: collapse; font-size: 0.85rem; }
th, td { padding: 8px 4px; text-align: center; border-bottom: 1px solid var(--border-color); }
}

th { background-color: #f1f5f9; position: sticky; top: 0; }

input.table-input {
  width: 80%;
  padding: 6px;
  border: 1px solid var(--border-color);
  border-radius: 4px;
  text-align: center;
}

.btn-group { display: flex; gap: 6px; margin-top: 10px; flex-wrap: wrap; }
button {
  background-color: var(--primary-color);
  color: white;
  border: none;
  padding: 8px 12px;
  border-radius: 6px;
  font-weight: 600;
  cursor: pointer;
  font-size: 0.85rem;
}
button:hover { background-color: var(--primary-hover); }
button.btn-secondary { background-color: #64748b; }
button.btn-danger { background-color: var(--accent-red); }
button.btn-audio { background-color: var(--accent-purple); padding: 4px 8px; font-size:
0.75rem; }

.notice-box {
  background-color: #fff6ff;
  border-left: 4px solid var(--primary-color);
  padding: 10px 12px;
  border-radius: 4px;
  font-size: 0.85rem;
  margin-bottom: 12px;
}

.equation-box {

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    background: #f8fafc;
    border: 1px solid #cbd5e1;
    padding: 10px;
    border-radius: 8px;
    text-align: center;
    font-weight: bold;
    margin-top: 12px;
    font-size: 0.95rem;
}

.chart-container { position: relative; height: 320px; width: 100%; }

.pitch-badge {
    background-color: #f3e8ff;
    color: #6b21a8;
    padding: 2px 6px;
    border-radius: 4px;
    font-weight: bold;
}

.control-panel {
    background-color: #f0fdf4;
    border: 1px solid #bbf7d0;
    padding: 12px;
    border-radius: 8px;
    margin-bottom: 12px;
}

.control-group {
    display: flex;
    align-items: center;
    gap: 10px;
    margin-top: 6px;
}

.control-group input[type="range"] {
    flex: 1;
}

.control-group input[type="number"] {
    width: 70px;
    padding: 4px;
    text-align: center;
    border: 1px solid var(--border-color);
    border-radius: 4px;
}
</style>
</head>

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<body>

<div class="container">
  <header>
    <h1>고등학교 과학탐구: 줄의 길이 및 진동수 조절 음높이 분석</h1>
    <p style="margin:0; color:#64748b; font-size:0.85rem;">기준 진동수와 줄의 길이를
    각각 자유롭게 조절하며 체감 음높이의 변화를 시뮬레이션합니다.</p>
  </header>

  <div class="grid-layout">
    <!-- 1. 데이터 입력 및 진동수 조절 -->
    <div class="card">
      <h2 style="font-size:1.1rem; margin-top:0;">1. 기준 진동수 조절 및 줄의 길이
      입력</h2>

      <!-- 진동수 직접 조절 패널 -->
      <div class="control-panel">
        <label style="font-weight:bold; font-size:0.85rem; color:#166534; display:block;">
          ⚙ 기준 진동수 조절 (20cm 기준 음높이)
        </label>
        <div class="control-group">
          <input type="range" id="baseFreqSlider" min="100" max="1000" value="440"
step="1" oninput="syncBaseFreq(this.value)">
          <input type="number" id="baseFreqInput" min="100" max="2000" value="440"
oninput="syncBaseFreq(this.value)">
          <span style="font-size:0.85rem; font-weight:bold;">Hz</span>
        </div>
      </div>

      <div class="manual-table-container">
        <table>
          <thead>
            <tr>
              <th>순번</th>
              <th>줄의 길이 L (cm)</th>
              <th>삭제</th>
            </tr>
          </thead>
          <tbody id="tableBody"></tbody>
        </table>
      </div>

      <div class="btn-group">
        <button type="button" id="btnAddRow">+ 길이 추가</button>
        <button type="button" class="btn-secondary" id="btnSample">샘플
        불러오기</button>
        <button type="button" class="btn-secondary" id="btnClear">초기화</button>
      </div>
    </div>
  </div>

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        <div style="margin-top: 12px;">
            <button type="button" id="btnAnalyze" style="width: 100%; padding: 10px;
font-size: 0.95rem;">분석 및 그래프 작성</button>
        </div>
    </div>

    <!-- 2. 결과 및 소리 듣기 -->
    <div class="card">
        <h2 style="font-size:1.1rem; margin-top:0;">2. 최종 계산된 체감 진동수 및
계이름</h2>
        <div id="noticeContainer">
            <div class="notice-box">길이를 입력 후 버튼을 눌러주세요.</div>
        </div>

        <h3 style="font-size:0.95rem; margin:8px 0;">길이별 출력 진동수 및 음고</h3>
        <div style="max-height: 180px; overflow-y: auto;">
            <table>
                <thead>
                    <tr>
                        <th>길이 L</th>
                        <th>출력 진동수</th>
                        <th>음의 높낮이</th>
                        <th>소리 듣기</th>
                    </tr>
                </thead>
                <tbody id="summaryTableBody">
                    <tr><td colspan="4" style="color:#94a3b8;">데이터가 없습니다.</td></tr>
                </tbody>
            </table>
        </div>

        <div id="equationDisplay" class="equation-box" style="display:none;">
            음높이 관계식: <span id="eqText"></span>
        </div>
    </div>

    <!-- 3. 차트 -->
    <div class="card">
        <h2 style="font-size:1.1rem; margin-top:0;">3. 줄의 길이에 따른 음의 높낮이(진동수)
관계 그래프</h2>
        <div class="chart-container">
            <canvas id="analysisChart"></canvas>
        </div>
    </div>
</div>

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<script>
let myChart = null;
let audioCtx = null;
let currentBaseFreq = 440; // 기본 기준 진동수
const BASE_LEN = 20;      // 기준 길이 (20cm)

function getAudioContext() {
  if (!audioCtx) {
    audioCtx = new (window.AudioContext || window.webkitAudioContext)();
  }
  if (audioCtx.state === 'suspended') {
    audioCtx.resume();
  }
  return audioCtx;
}

function playTone(freq) {
  if (!freq || freq <= 0) return;
  try {
    const ctx = getAudioContext();
    const osc = ctx.createOscillator();
    const gain = ctx.createGain();

    osc.type = 'sine';
    osc.frequency.setValueAtTime(freq, ctx.currentTime);

    gain.gain.setValueAtTime(0.3, ctx.currentTime);
    gain.gain.exponentialRampToValueAtTime(0.0001, ctx.currentTime + 0.8);

    osc.connect(gain);
    gain.connect(ctx.destination);

    osc.start();
    osc.stop(ctx.currentTime + 0.8);
  } catch (e) {
    console.error("Audio error:", e);
  }
}

// 기준 진동수 조절 동기화 함수
function syncBaseFreq(val) {
  const num = parseFloat(val);
  if (!isNaN(num) && num > 0) {
    currentBaseFreq = num;
    document.getElementById('baseFreqSlider').value = num;
    document.getElementById('baseFreqInput').value = num;
    processData();
  }
}

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}

function freqToPitch(freq) {
  if (!freq || freq <= 0) return "-";
  const noteNames = ["도 (C)", "도#", "레 (D)", "레#", "미 (E)", "파 (F)", "파#", "솔 (G)",
    "솔#", "라 (A)", "라#", "시 (B)"];

  const midi = Math.round(69 + 12 * Math.log2(freq / 440));
  const noteIndex = (midi % 12 + 12) % 12;
  const octave = Math.floor(midi / 12) - 1;

  return `${noteNames[noteIndex]}${octave}`;
}

document.addEventListener('DOMContentLoaded', function() {
  document.getElementById('btnAddRow').addEventListener('click', () => addRow());
  document.getElementById('btnSample').addEventListener('click', loadSampleData);
  document.getElementById('btnClear').addEventListener('click', clearTable);
  document.getElementById('btnAnalyze').addEventListener('click', processData);

  loadSampleData();
});

function addRow(len = "") {
  const tbody = document.getElementById('tableBody');
  const tr = document.createElement('tr');

  tr.innerHTML = `
    <td class="row-index"></td>
    <td><input type="number" step="any" class="table-input len-input" value="${len}"
placeholder="예: 10, 20, 30" oninput="processData()"></td>
    <td><button type="button" class="btn-danger" style="padding: 2px 6px;"
onclick="removeRow(this)">X</button></td>
  `;

  tbody.appendChild(tr);
  updateRowNumbers();
}

function removeRow(btn) {
  btn.closest('tr').remove();
  updateRowNumbers();
  processData();
}

function updateRowNumbers() {
  const rows = document.querySelectorAll('#tableBody tr');
  rows.forEach((row, idx) => {

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        const indexCell = row.querySelector('.row-index');
        if (indexCell) indexCell.innerText = idx + 1;
    });
}

function clearTable() {
    document.getElementById('tableBody').innerHTML = "";
    addRow();
    processData();
}

function loadSampleData() {
    document.getElementById('tableBody').innerHTML = "";
    const sampleLengths = [10, 15, 20, 30, 40, 50];
    sampleLengths.forEach(l => addRow(l));
    processData();
}

function processData() {
    const rows = document.querySelectorAll('#tableBody tr');
    let validData = [];
    let excludedCount = 0;

    rows.forEach(row => {
        const lenInput = row.querySelector('.len-input');
        if (!lenInput) return;

        const lenVal = lenInput.value.trim();

        if (lenVal === "" || isNaN(lenVal)) {
            excludedCount++;
        } else {
            const L = parseFloat(lenVal);
            if (L <= 0) {
                excludedCount++;
            } else {
                // 조절된 기준 진동수에 따른 체감 진동수 계산 (f = f_base * (20 / L))
                const actualFreq = currentBaseFreq * (BASE_LEN / L);
                validData.push({ L, f: actualFreq });
            }
        }
    });
}

const noticeContainer = document.getElementById('noticeContainer');
if (validData.length === 0) {
    noticeContainer.innerHTML = `<div class="notice-box"
style="border-color:var(--accent-red); color:var(--accent-red);">유효한 길이 데이터가
없습니다.</div>`;
}

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    return;
}

noticeContainer.innerHTML = `
    <div class="notice-box">
        <strong>기준 진동수 ${currentBaseFreq}Hz 적용 완료:</strong> 총
        ${validData.length}개 위치의 음높이가 계산되었습니다.
    </div>
`;

const summaryTbody = document.getElementById('summaryTableBody');
summaryTbody.innerHTML = "";

validData.sort((a, b) => a.L - b.L);

validData.forEach(item => {
    const pitchText = freqToPitch(item.f);
    summaryTbody.innerHTML += `
        <tr>
            <td><strong>${item.L} cm</strong></td>
            <td><strong>${item.f.toFixed(1)} Hz</strong></td>
            <td><span class="pitch-badge">${pitchText}</span></td>
            <td><button type="button" class="btn-audio" onclick="playTone(${item.f})">🔊
소리 듣기</button></td>
        </tr>
    `;
});

const k = currentBaseFreq * BASE_LEN;
document.getElementById('equationDisplay').style.display = 'block';
document.getElementById('eqText').innerText = `f = ${k.toFixed(0)} / L (기준
${currentBaseFreq}Hz 적용됨)`;

renderChart(validData, k);
}

function renderChart(dataPoints, k) {
    const ctx = document.getElementById('analysisChart').getContext('2d');
    if (myChart) myChart.destroy();

    const minL = Math.min(...dataPoints.map(p => p.L));
    const maxL = Math.max(...dataPoints.map(p => p.L));

    const trendlineData = [];
    const startL = Math.max(1, minL * 0.8);
    const endL = maxL * 1.1;
    const step = (endL - startL) / 100;

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for (let l = startL; l <= endL; l += step) {
  trendlineData.push({ x: l, y: k / l });
}

myChart = new Chart(ctx, {
  type: 'scatter',
  data: {
    datasets: [
      {
        label: '측정 위치 음높이 (Hz)',
        data: dataPoints.map(p => ({ x: p.L, y: p.f })),
        backgroundColor: 'rgba(139, 92, 246, 0.8)',
        pointRadius: 7
      },
      {
        type: 'line',
        label: `반비례 곡선 ( $k = ${k.toFixed(0)}$ )`,
        data: trendlineData,
        borderColor: 'rgba(37, 99, 235, 0.8)',
        borderWidth: 2,
        fill: false,
        pointRadius: 0
      }
    ]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    scales: {
      x: { title: { display: true, text: '줄의 길이 L (cm)' } },
      y: { title: { display: true, text: '실제 발생 진동수 f (Hz)' } }
    }
  }
});
}
</script>

</body>
</html>

```